How to Restructure Data in SPSS

STC224

Introduction

- When running repeated measures of data, it is necessary to organize the data so that each row represents an individual case.
- Sometimes in the raw data, the rows will be organized as individual measurements for the same cases.
- The example below represents two participants with two outcome measurements (weight and calories), measured at three different time points.
- There are two formats for representing repeated measures data:
  - **the long and wide format.**

The long format.

- The long format utilizes multiple rows for each observation:

ID	WEIGHT	CALORIES	TIME
1	235	3000	1
1	205	2900	2
1	160	2800	3
2	155	2950	1
2	150	2900	2
2	140	2800	3

The wide format.

- The wide format utilizes one row for each observation or participant:

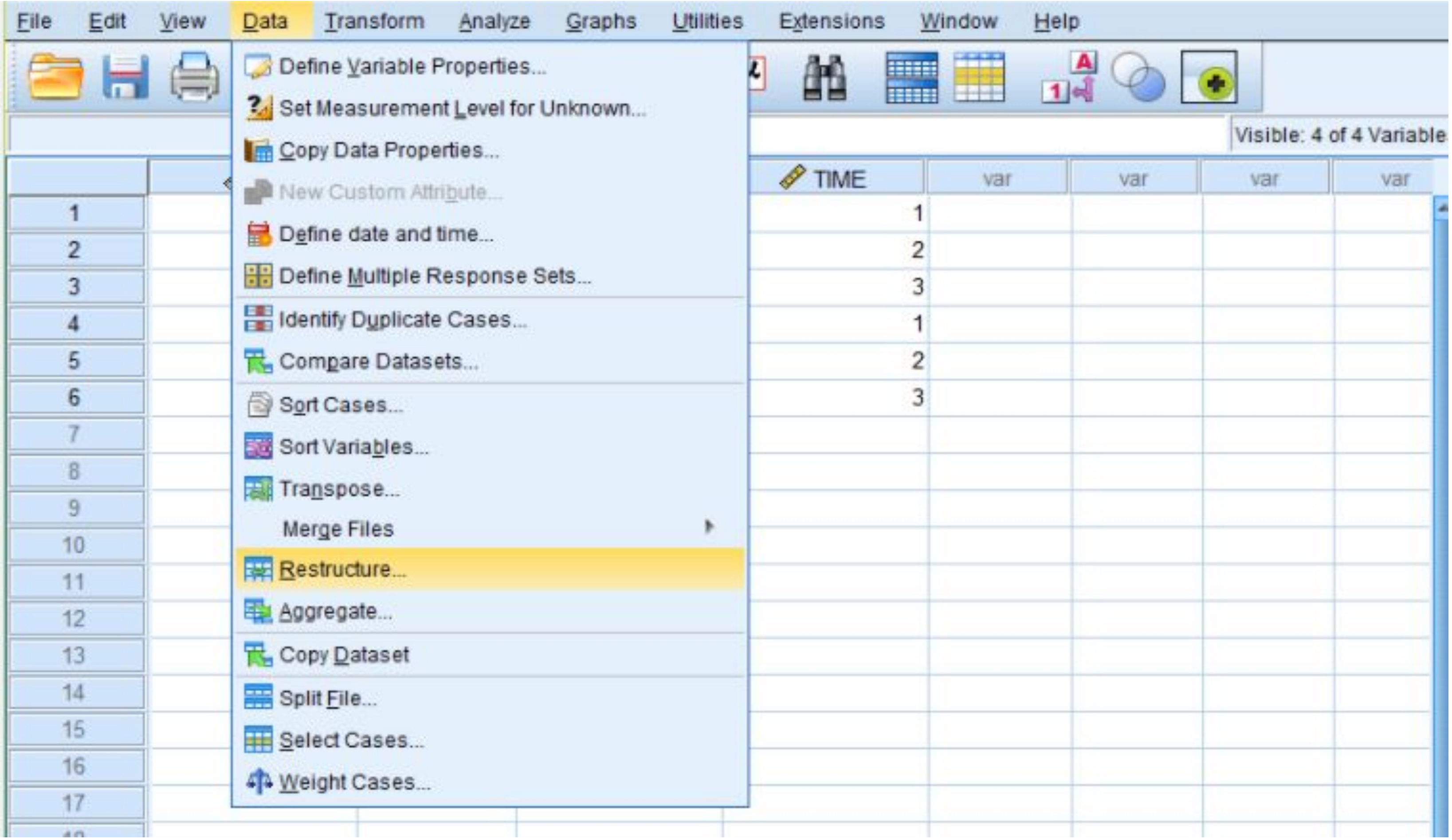
ID	weight1	weight2	weight3	calories1	calories2	calories3
1	235	205	160	3000	2900	2800
2	155	150	140	2950	2900	2800

Restructuring using SPSS

- Using SPSS, the data can be restructured from long format into wide format.
- Here we present the steps for doing this using the above example data.

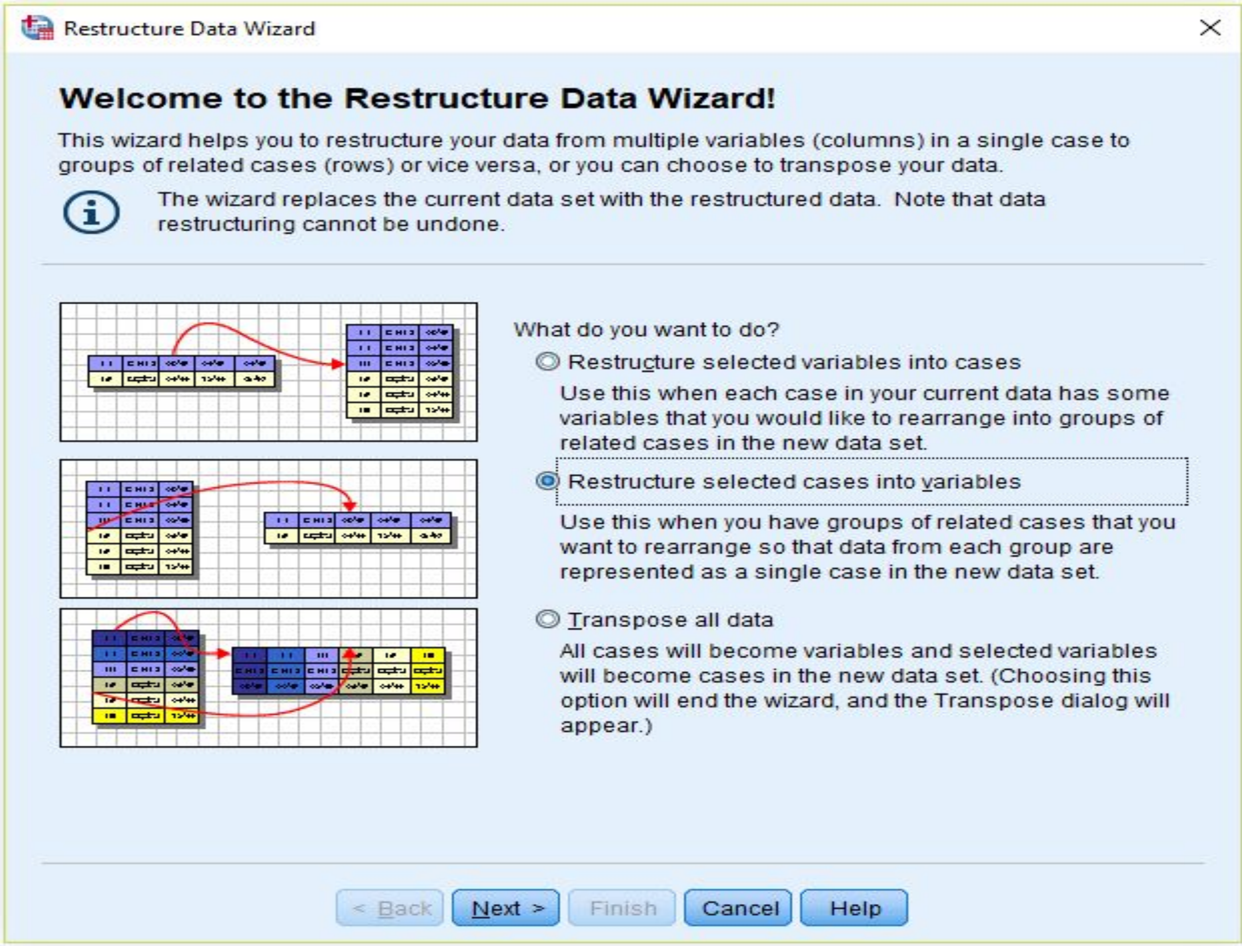
Restructuring using SPSS cont'd

1. From the Data menu, select Restructure…



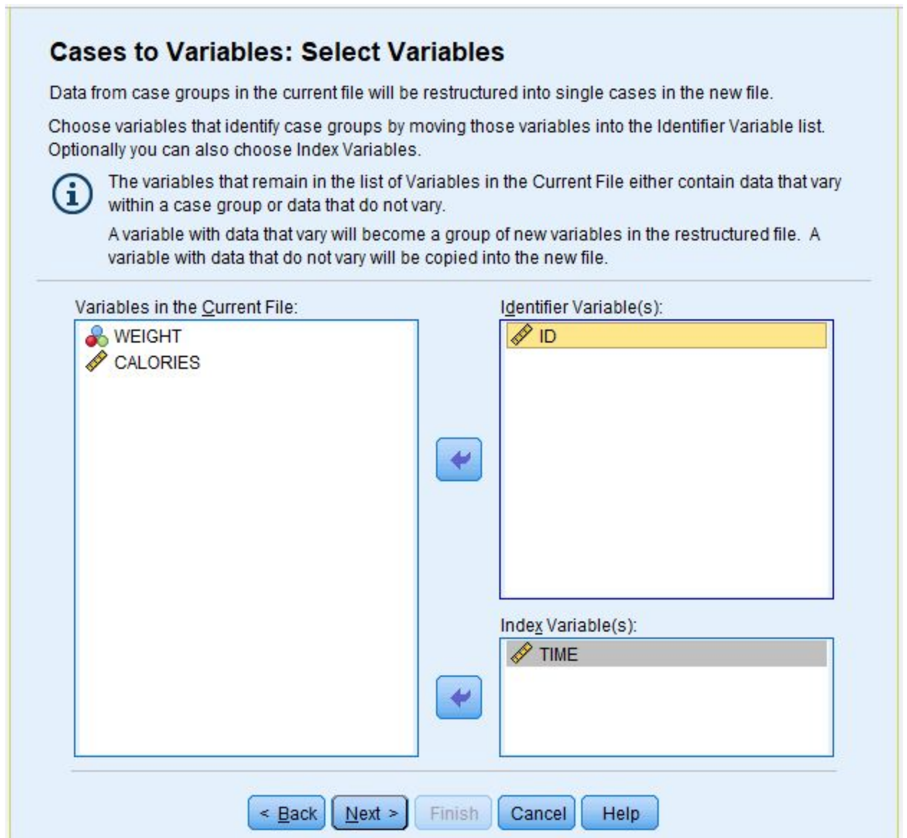
Restructuring using SPSS cont'd

2. Then select "Restructure selected cases into variables".



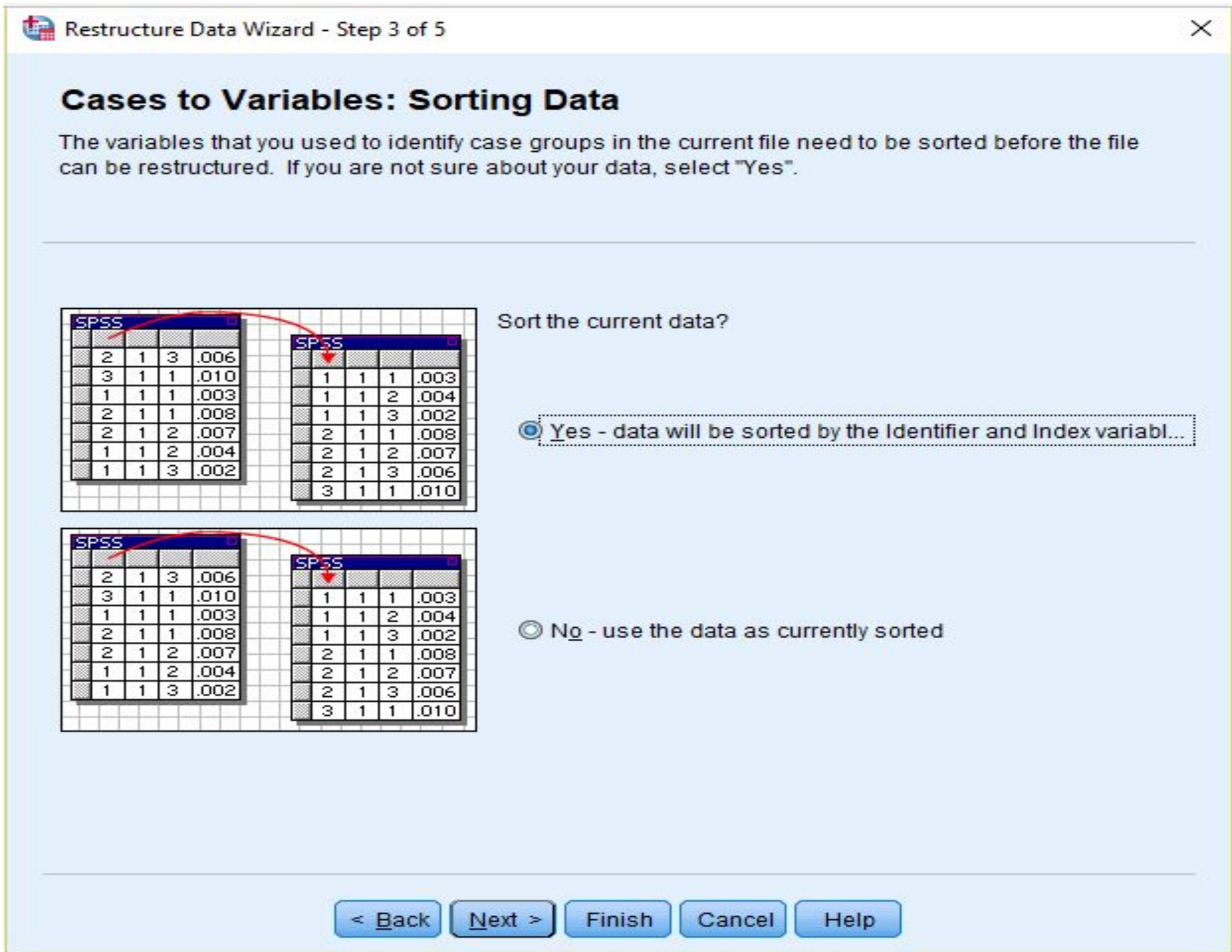
Restructuring using SPSS cont’d

3. For the “Identifier Variable(s)” box, transfer over the ID variable. For “Index Variable(s)”, enter the designator for the repeated measurement (in this case, it is TIME). Then click Next.



Restructuring using SPSS cont’d

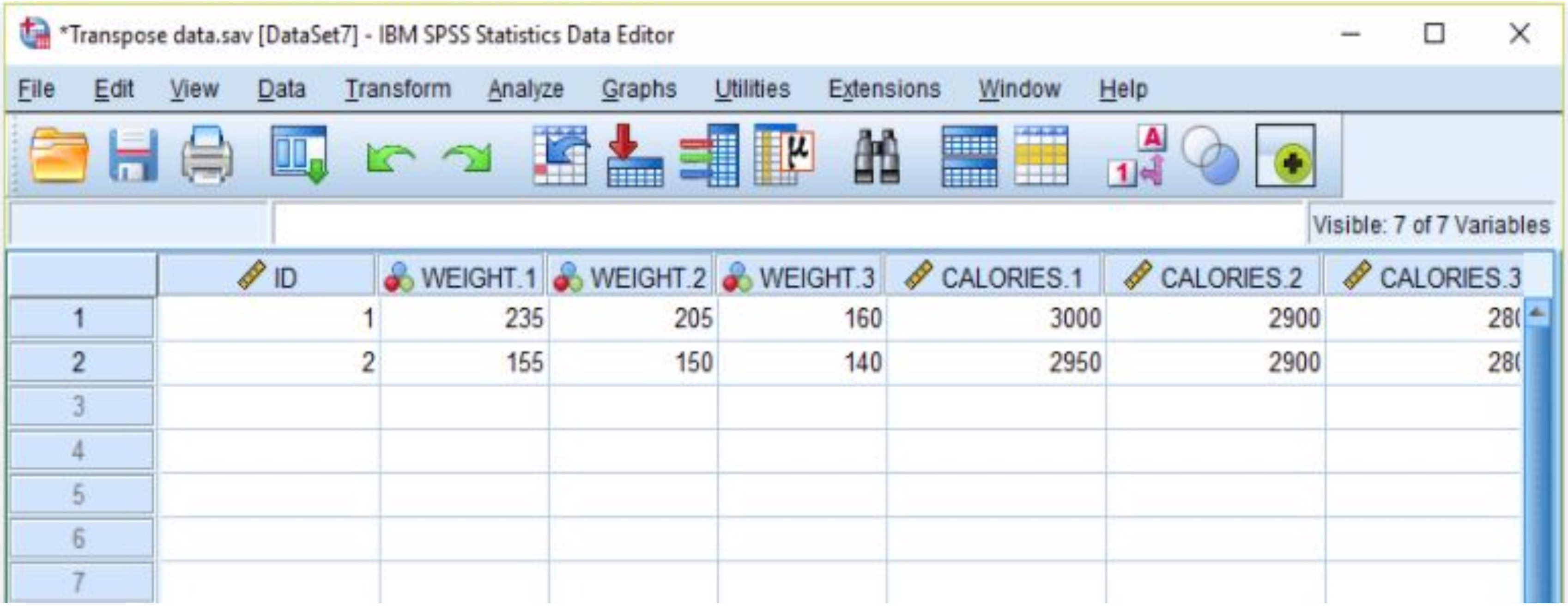
4. Select “Yes” to “Sort the current data”.



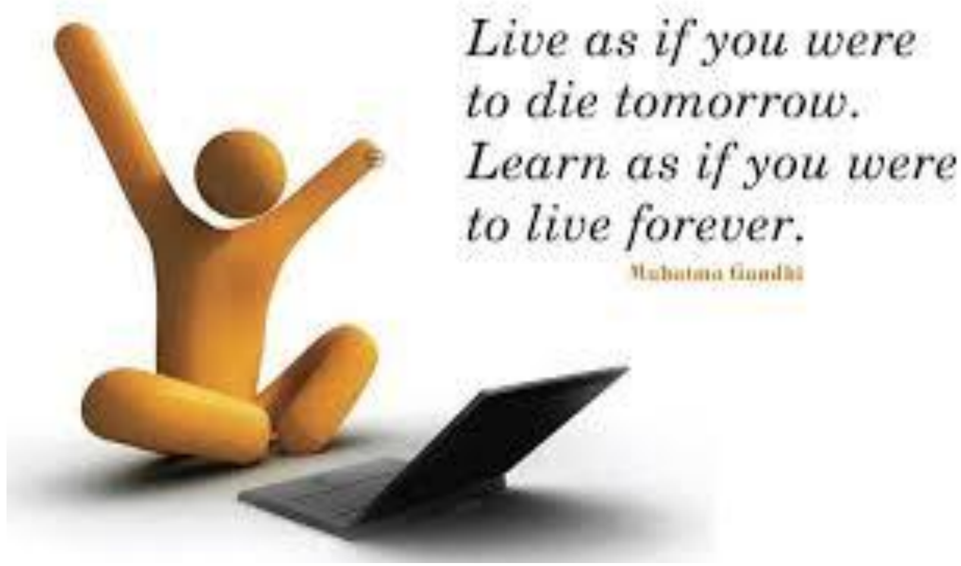
Restructuring using SPSS cont’d

5. Click Finish and the data should be restructured to a wide format. Save the file as a transposed version.
   - Note how the variables are automatically given a different name based on the number of measurements.
   - With the data in this format, you can now properly conduct repeated-measures analyses (such as dependent samples *t*-tests or repeated-measures ANOVAs).

Restructuring using SPSS cont’d



	ID	WEIGHT.1	WEIGHT.2	WEIGHT.3	CALORIES.1	CALORIES.2	CALORIES.3
1	1	235	205	160	3000	2900	28
2	2	155	150	140	2950	2900	28
3							
4							
5							
6							
7							


Live as if you were
to die tomorrow.
Learn as if you were
to live forever.